

will produce also depends on the image itself. For example, an image that is already pixelated will look very bad even if its file size is reduced by a small amount, while a photo from a good digital camera can be reduced to up to 60% of its original file size with no or hardly any visible loss in quality. However, assuming a similar quality of the original images, a rule of thumb is that those that have high detail—that is, considerable variation of colours across the image—must not be reduced much (say, less than 80%), while images that have uniform spreading of a single colour can be reduced to about 40% or lesser of their original size.

2.2. Changing Resolution / Pixels (Physical Image Size)

Of all digital imaging concepts, *resolution* is probably the least understood. With conventional film cameras, you just clicked the photo and relied on a developing shop (or *lab* as they called them) to give you the photo. In the world of digital photography, however, you are faced with resolution, dpi, image size, etc. if you are going to print it yourself. Digital imaging suddenly becomes simpler the moment you understand what resolution and dpi are all about. Let us look at the concept of resolution and pixels taking an oft-used analogy. We haven't invented this analogy, and are certainly not the only ones to have used it!

Imagine a large playground, analogous to your screen or an image. Now let's say there are a large number of children wearing hats of a single colour—say blue, standing in 100 rows and columns, close to each other. If you look at this arrangement from a sufficient height, the children will appear as a square of blue. If the rows and columns are increased by another 100, then you will see a larger image. Similarly, if you have a row and column of only 50 x 50, you'll see a smaller blue square.

Now think of each individual hat as a pixel. This is the most basic element that an image is composed of. The number of chil-